

A BAYESIAN MODEL FOR ESTIMATING THE FAILURE RATE FOR DIFFERENT GROUPS

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Summary: Gamerman (1991) used the dynamic generalized modelling approach of West, Harrison and Migon (1985) to estimate the Cox type hazard function. Lindley and Singpurwalla (1986) introduced an environmental parameter for constructing dependence between lifetimes. In this paper the estimation of the failure rate from dependent lifetimes of components operating under similar conditions in different groups is considered using the dynamic generalized linear model approach and the introduction of an environmental parameter. Illustrations of the procedure, using real data, are given.

1. INTRODUCTION

Gamerman (1991) introduced a dynamic linear Bayesian model for estimating the hazard on a set of lifetime data where the same type of item functions under different treatments. His approach was based on the Bayesian approach to modelling presented by West, Harrison and Migon (1985) and the analysis is such that information collected up to a time is passed sequentially through intervals, since information gathered in an interval should exert some influence on the adjacent intervals. A motivation for using the dynamic model (Gamerman, 1991) is that this type of model allows the effects of covariates to vary in time through a stochastic evolution and therefore the form of variation is determined by the data and prior assumptions. The form of prior is arbitrary and values may be chosen to incorporate subjective information or to avoid numerical difficulties. Thus,

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dynamic models provide a natural way of blending subjective information and data. Lindley and Singpurwalla (1986) introduced an environmental parameter in the failure rates of two different types of components to bring in a dependence between the lifetimes of the components. In considering the multi-group case, we used the idea of an environmental parameter to induce dependence between the joint lifetimes which describe the system and then apply the technique of the dynamic generalized Bayes model to estimate the failure rates or hazards.

In section 2 we consider the dynamic linear Bayes model on a single failure time variable with the inclusion of an environmental parameter and where distributional assumptions have to be made regarding the environment in which items function. The approach is applied in a numerical example in section 3.

In section 4 we extend the results to the case where failure times are considered for different groups. This is a case where a system consists of different types of items (components, individuals) which operate under certain stresses (conditions) and the common environment has an effect on the lifetimes of items. An illustration is given in section 5 by using real data.

2. THE SINGLE-GROUP CASE

Assume the lifetime T of a component is piecewise exponentially distributed in the interval $I_n = (t_{n-1}, t_n]$, $n = 1, 2, \dots, N$, conditional on the failure rate λ , which is specified on the basis of controlled life tests on components, and an environmental effect which can increase or decrease the failure rate by the factor η . The conditional density function for the current interval is given by

$$(1) \quad p(t | \eta, \lambda_n) = \eta \lambda_n e^{-\eta \lambda_n (t - t_{n-1})}, \quad t \in I_n.$$

Consideration of this piecewise exponential (PE) distribution for observations are sufficiently flexible to constitute a good practical approximation to any continuous distribution (Gamerman, 1991). The use

of PE distributions in survival data analysis was also proposed by Breslow (1972, 1974).

From (1) it can be seen that the hazard function is given by

$$h(t) = \eta \lambda_n$$

and assume that

$$\eta = e^{\beta_0} \quad \text{and}$$

$$\lambda_n = e^{z\beta_n^{(1)'}}$$

$$z = (z_1, \dots, z_p), \quad \beta_n^{(1)} = (\beta_1, \dots, \beta_p).$$

z_1 may be taken as one and $z_2, \dots, z_p$ may indicate different stresses under which different components of the same type are operating. The guide relation is given by

$$(2) \quad \ln \eta \lambda_n = \beta_0 + z\beta_n^{(1)'}$$

and the way in which the regression parameter vector $\beta_n = (\beta_0 \quad \beta_n^{(1)'})$ (system evolution on β_n) may vary from interval I_{n-1} to I_n is

$$(3) \quad \beta_n = G_n \beta_{n-1} + w_n$$

where G_n is the system evolution matrix and w_n is the evolution error of β_n over the interval. Normality is not assumed, only the first- and second-order moments of the evolution error, namely $E(w_n) = 0$ and $\text{Var}(w_n) = W_n$ being so far assumed.

Evolution is therefore controlled by the variance which may be chosen to allow for a range of behaviours.

By following the dynamic generalized Bayes approach introduced by West, Harrison and Migon (1985), we proceed to develop an approximate analysis based on assuming partially specified distributions in terms of first- and second-order moments, i.e. assume that the posterior distribution of β_{n-1} is partially specified with mean m_{n-1} and variance C_{n-1} , denoted by

$$\beta_{n-1} | D_{n-1} \sim [m_{n-1}, C_{n-1}],$$

where D is the set containing the information thus far obtained.

It follows that, from (3), the prior moments for the next interval parameter β_n are

$$\beta_n | D_{n-1} \sim [a_n, P_n]$$

where

$$a_n = G_n m_{n-1}, \quad P_n = G_n C_{n-1} G'_n + W_n.$$

The likelihood for an individual, given information D , is then

$$(4) \quad l(\eta, \lambda) = (\eta\lambda)^\delta \exp(-\eta\lambda(t - t_{n-1}))$$

$$\text{where } \delta = \delta_n^{(j)} = \begin{cases} 1 & \text{if a failure is observed} \\ 0 & \text{if a component is still functioning or censoring has occurred.} \end{cases}$$

$\lambda = \lambda_n^{(j)}$ since it is estimated after each observation. It is assumed that a change in environment has a similar effect on the lifetimes of all components. Stress may, for example, be regarded as the different temperatures under which components of the same type are functioning in a system, while the environmental effect may be the vibration of the system which will have the same effect on all components (for example, halves the expected lifetime of every component). Sub- and superscripts will be omitted for convenience of notation.

β and $\ln \eta\lambda$ have joint prior distributions that are only partially specified in terms of moments.

$$(5) \quad \begin{pmatrix} \beta \\ \ln \eta\lambda \end{pmatrix} \sim \begin{bmatrix} a \\ f \end{bmatrix}, \begin{pmatrix} P & s \\ s' & q \end{pmatrix}$$

where

$$\beta = \begin{pmatrix} \beta_0 \\ \beta^{(1)} \end{pmatrix}, \quad a = \begin{pmatrix} a_0 \\ a^{(1)} \end{pmatrix}, \quad P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}$$

Given information D (it can either be the observance of a failure time t or the end of the interval I_n), the posterior moments of β from (5) can then be obtained by using linear Bayes methods:

$$(6) \quad m = E(\beta | D) = a + \frac{s'}{q} [E(\ell n \eta \lambda | D) - f]$$

and

$$(7) \quad C = \text{Var}(\beta | D) = P - \frac{ss'}{q} + \frac{ss'}{q^2} \text{Var}(\ell n \eta \lambda | D)$$

In order to obtain $E(\ell n \eta \lambda | D)$ and $\text{Var}(\ell n \eta \lambda | D)$, further assumptions about the form of prior for η and λ are necessary. In any Bayesian analysis it is very important to give a good justification of the prior distribution considered. It is, however, important to note that in this case the form of the prior is arbitrary (as stated by West et al., 1985); apart from $(\ell n \eta \lambda | D) \sim [f; q]$, no further restrictions have been made on the prior. Values may be chosen to incorporate subjective information or to avoid numerical difficulties. Since the operating environment is to either increase or decrease the failure rate of a component, a reasonably flexible and analytically tractable distribution to describe uncertainty about the unknown environment parameter η , also adopted by Lindley and Singpurwalla (1986), is the Gamma distribution. By considering the joint prior on η and λ to be a Gamma-Beta type 2 distribution, $GB_2(\alpha, \tau, \rho, \gamma)$, with density function

$$(8) \quad p(\eta, \lambda) \propto \eta^{\alpha-1} e^{-\tau\eta} \lambda^{\rho-1} e^{-\gamma\lambda}, \quad \eta, \lambda > 0$$

and proportionality constant given by

$$\frac{\tau^{\alpha-\rho} \gamma^\rho}{\Gamma(\rho) \Gamma(\alpha - \rho)},$$

the marginal densities of η and λ are found to be respectively

$$f(\eta) = \frac{\tau^{\alpha-\rho}}{\Gamma(\alpha - \rho)} \eta^{\alpha-\rho-1} e^{-\tau\eta}, \quad \eta > 0$$

and

$$f(\lambda) = \frac{(\gamma/\tau)^\rho}{B(\rho, \alpha - \rho)} \lambda^{\rho-1} (1 + \frac{\gamma}{\tau} \lambda)^{-\alpha}, \quad \lambda > 0.$$

Thus, $\eta \sim G(\alpha - \rho, \tau)$, the Gamma distribution, and $\lambda \sim B_2(\rho, \alpha - \rho, \frac{\tau}{\tau})$, the Beta type 2 distribution.

By combining the likelihood (4) with the prior (8) the posterior distribution is found to be $\eta, \lambda | D \sim GB_2(\alpha + \delta, \tau, \rho + \delta, \gamma + t - t_{n-1})$, with density function

$$(9) \quad p(\eta, \lambda | D) \propto \eta^{\alpha + \delta - 1} e^{-\tau \eta} \lambda^{\rho + \delta - 1} e^{-(\gamma + t - t_{n-1}) \eta \lambda}, \quad \eta, \lambda > 0.$$

The proportionality constant is $\frac{\tau^{\alpha - \rho} (\gamma + t - t_{n-1})^{\rho + \delta}}{\Gamma(\rho + \delta) \Gamma(\alpha - \rho)}$ and the marginal posterior distributions are $\eta | D \sim G(\alpha - \rho, \tau)$ and $\lambda | D \sim B_2 \left[\rho + \delta, \alpha - \rho, \frac{\gamma + t - t_{n-1}}{\tau} \right]$,

$$\text{with} \quad E(\eta | D) = \frac{\alpha - \rho}{\tau}$$

$$\text{Var}(\eta | D) = \frac{\alpha - \rho}{\tau^2}, \quad \tau > 0$$

$$E(\gamma | D) = \frac{\tau(\rho + \delta)}{(\gamma + t - t_{n-1})(\alpha - \rho - 1)}$$

$$\text{Var}(\gamma | D) = \frac{\tau^2(\rho + \delta)(\alpha + \delta - 1)}{(\gamma + t - t_{n-1})^2(\alpha - \rho - 1)^2(\alpha - \rho - 2)}, \quad \gamma > 0, \alpha - \rho > 2$$

and

$$\text{Cov}(\eta, \lambda | D) = \frac{\rho + \delta}{(\gamma + t - t_{n-1})(\alpha - \rho - 1)}.$$

A Taylor series expansion will be used to approximate $E(\ell n \eta \lambda | D)$ and $\text{Var}(\ell n \eta \lambda | D)$ in (6) and (7) respectively:

$$E(\ell n \eta \lambda | D) \simeq \ell n(\mu_\eta \mu_\lambda), \quad \text{where } \mu_\eta = E(\eta | D) \text{ and } \mu_\lambda = E(\lambda | D).$$

$$\text{Var}(\ell n \eta \lambda | D) \simeq \sigma_\eta^2 [g'(\mu_\eta)]^2 + \sigma_\lambda^2 [g'(\mu_\lambda)]^2 + 2\sigma_{\eta\lambda} [g'(\mu_\eta)g'(\mu_\lambda)],$$

where

$$\sigma_\eta^2 = \text{Var}(\eta | D), \quad \sigma_\lambda^2 = \text{Var}(\lambda | D)$$

$$g'(\mu_\eta) = \left. \frac{\partial \ell n(\eta \lambda)}{\partial \eta} \right|_{\eta = \mu_\eta}, \quad g'(\mu_\lambda) = \left. \frac{\partial \ell n(\eta \lambda)}{\partial \lambda} \right|_{\lambda = \mu_\lambda}$$

and

$$\sigma_{\eta\lambda} = \text{Cov}(\eta, \lambda | D).$$

It follows that

$$(10) \quad E(\ln \eta \lambda | D) = \ln \frac{(\alpha - \rho)(\rho + \delta)}{(\alpha - \rho - 1)(\gamma + t - t_{n-1})}$$

$$(11) \quad \text{Var}(\ln \eta \lambda | D) = \frac{(\alpha + \delta - 1)}{(\rho + \delta)(\alpha - \rho - 2)} - \frac{1}{(\alpha - \rho)}.$$

The prior parameters α, ρ, τ and γ can be expressed in terms of the moments given in (5) by equating the moments from $GB_2(\alpha, \tau, \rho, \gamma)$ to that given in (5) and using the guide relations

$$\ln \eta = \beta_0, \ln \lambda = z\beta^{(1)'}$$

Solving the four equations $E(\ln \eta) = \alpha_0$, $E(\ln \lambda) = za^{(1)'}$, $\text{Var}(\ln \eta) = p_{00}$, $\text{Var}(\ln \lambda) = zP_{11}z'$, for the four parameters α, τ, ρ and γ , we obtain

$$(12) \quad \alpha = \frac{p_{00}^2 - zP_{11}z'(1 - 2p_{00})}{p_{00}^2 - p_{00}zP_{11}z'(1 - 2p_{00})}$$

$$\rho = \frac{p_{00} - 1}{p_{00} - zP_{11}z'(1 - 2p_{00})}$$

$$\gamma = \frac{e^{-f}}{zP_{11}z'(1 - 2p_{00}) - p_{00}}$$

$$\tau = p_{00}^{-1} e^{-a_0}, \text{ where } f = a_0 + za^{(1)'}$$

Substituting these solutions in (10) and (11) we obtain the posterior moments of β given in (6) and (7). Let $a = G_n m$ and $P = G_n CG_n' + W_n$ and continue updating the posterior moments for each individual (component)

known to operate at the beginning of the interval I_n with t the failure time if it fails in I_n , otherwise $t = t_n$.

Repeat the procedure for all intervals I_n , $n = 1, \dots, N$.

As information is processed, the on-line distributions for $(\beta_n | D)$, $n = 1, 2, \dots, N$, become available. The distributions for the parameters given all information gathered, D_N , are, however, useful as stated by Gamerman (1991). These smoothed distributions are obtained through a recursive algorithm used in standard dynamic linear models and can be extended to DGLMs, for which the proof can be found in Gamerman (1991).

3. ILLUSTRATION

Table 1 gives the ordered lifetimes (in hours) of Kevlar 49/epoxy pressure vessels tested at 86% (4300 p.s.i.) and 80% (4000 p.s.i.) stress levels.

Table 1

Stress level	Lifetimes (in hours)							
86%	2.2	4.0	4.0	4.6	6.1	6.7	7.9	8.3
	8.5	9.1	10.2	12.5	13.3	14.0	14.6	15.0
	18.7	22.1	45.9	55.4	61.2	87.5	98.2	101.0
	111.4	144.0	158.7	243.9	254.1	444.4	590.4	638.2
	755.2	952.2	1108.2	1148.5	1569.3	1750.6	1802.1	
80%	19.1	24.3	69.8	71.2	136.0	199.1	403.7	
	432.2	453.4	514.1	514.2	541.6	544.9	554.2	
	664.5	694.1	876.7	930.4	1254.9	1275.6	1536.8	
	1755.5	2046.2	6177.5					

Barlow, Toland and Freeman (1988) assumed a Weibull life distribution model on the lifetimes of the pressure vessels with hazard function

$$h(t; \alpha, \beta) = \alpha\beta(\beta t)^{\alpha-1}, \quad \alpha, \beta > 0, \quad t > 0.$$

(For $\alpha = 1$, the distribution is exponential).

Table 2 shows the maximum likelihood estimates of α and β as calculated by Barlow et al. (1988).

Table 2

Stress level	α	β
86%	6.104×10^{-3}	0.520
80%	1.200×10^{-3}	0.875

To apply the model on the lifetimes of the pressure vessels, the following initial values for the moments of the prior distribution of β were used:

$$a_{00} = (0.7 \quad 0 \quad 0)', \quad P_{00} = \begin{bmatrix} 0.001 & 0 \\ 0 & 100I_2 \end{bmatrix}, \quad G_n = I_3$$

$$w_n = (0 \quad 0 \quad -0.05)', \quad W_n = \begin{bmatrix} 0.001 & 0 \\ 0 & 0.05I_2 \end{bmatrix}$$

The first element of the vector a_{00} would be close to the expected value of η , which is in this case $\exp(0.7) \approx 2$. By setting the first element of $a_{00} = 0.7$, parameters are estimated with the effect of the environment included. By using the dynamic model to determine the hazard, this environmental effect can also be omitted to give the hazard function under conditions where the lifetimes of components are expected to be double that of the lifetimes under the current environment. The initial value can therefore be changed to examine different hypothetical situations of the environment. The (1,1)-th element of P_{00} indicates a small variance on the environmental parameter η . To express prior ignorance (lack of information) the

other elements of a_{00} may be chosen as 0 and those on the diagonal of P_{00} as k , where k is large (Fahrmeier, 1994, Harvey, 1989), to indicate that only data information is used. For this example $k = 100$.

In practice, the intervals, I_n , should be chosen according to the expectation of noticeable changes in the intensity rate and the amount of time variation can be controlled by w_n . The values for W_n and w_n were chosen on an illustrative basis and can be changed for more flexibility to the inference. The time grid is a combination of equal length intervals, for this example $\{0, 120, 360, 600, \dots, 2040, 2520, \dots, 6360\}$, although observed failure times may also be used.

Data can also be analysed with an informative prior distribution set to coincide with data information. Examples shown in Gamerman (1991) provide basic guidance on the setting of prior moments, time grid and system evolution in practical problems.

The vector z is taken as $(1 \ 1 \ 0)$ or $(1 \ 1 \ 1)$, where the last element of the vector indicates whether the component is functioning under an 80% (0) or 86% (1) stress level. After substitution of the initial values in the model, the parameters α , τ , ρ and γ can be obtained and substitution of these solutions in (10) and (11) enable us to calculate the posterior moments of β , given by (6) and (7). After each observation, set $a_{nj+1} = m_{nj}$ and $P_{nj+1} = C_{nj}$ and move on to the next observation until all the observations known to be functioning at the beginning of I_n , say r_n , have been processed. Then, $D_{n-1;r_n} = D_n$, $m_{nr_n} = m_n$ and $C_{nr_n} = C_n$. This procedure is repeated for all intervals I_n . The smoothed estimates, a_n , of β_n obtained at the end of each interval are given in table 3.

In the graphs that follow the terms "hazard (BTF)" and "dynamic hazard" will be used. The term "hazard (BTF)" is referring to the distribution involved, with the parameters given in table 2 (Barlow, Toland and Freeman, 1988), while the term "dynamic hazard" is the hazard function obtained by applying the dynamic Bayes procedure.

Table 3

a_1	a_2	a_3	a_4	a_5
0.7008	0.6933	0.6911	0.6913	0.6909
-6.8381	-7.2402	-7.3696	-7.3695	-7.3981
1.1904	0.7223	0.4871	0.2680	0.1808
a_6	a_7	a_8	a_9	a_{10}
0.6911	0.6918	0.6912	0.6917	0.6905
-7.3974	-7.3639	-7.3984	-7.3790	-7.4377
0.1218	0.1177	0.0869	0.1196	0.1004
a_{11}	a_{12}	a_{13}	a_{14}	a_{15}
0.6888	0.6864	0.6846	0.6833	0.6825
-7.5264	-7.6481	-7.7394	-7.8042	-7.8456
0.0325	-0.0224	-0.0670	-0.1027	-0.1308
a_{16}	a_{17}	a_{18}	a_{19}	
0.6821	0.6821	0.6824	0.6830	
-7.8662	-7.8680	-7.8529	-7.8221	
-0.1521	-0.1673	-0.1770	-0.1817	

The dynamic (after smoothing) as well as the hazard (BTF) functions for the lifetimes of components functioning under an 80% and 86% stress, can be seen in figure 1. In figure 2 the smoothed dynamic hazard function can be seen for components functioning under an 86% stress where the effect of the environment is assumed present and where it is omitted, i.e. where the lifetimes of components are expected to be double that of the lifetimes under the current conditions. The hazard (BTF) is also shown on the graph. Figure 3 shows the same hazard functions as in figure 2, but for a stress level of 86%.

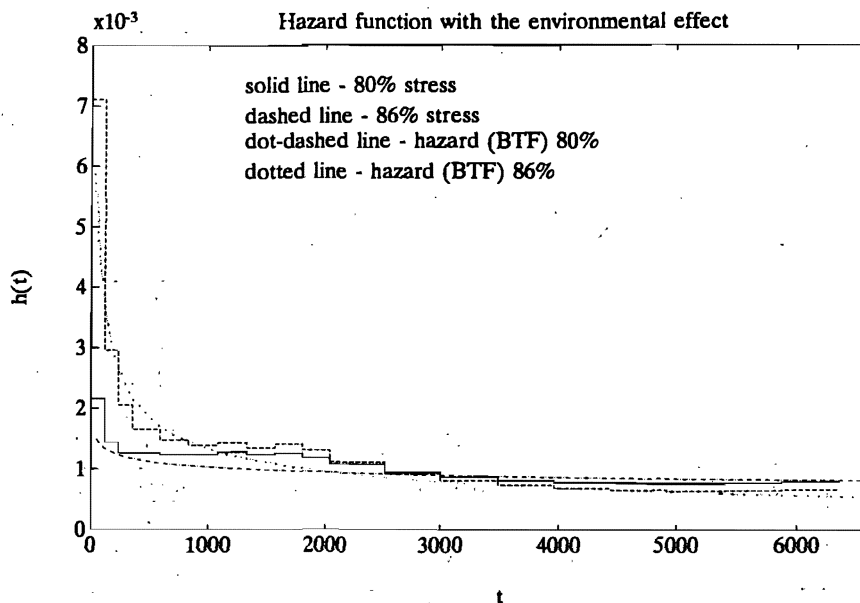


Figure 1

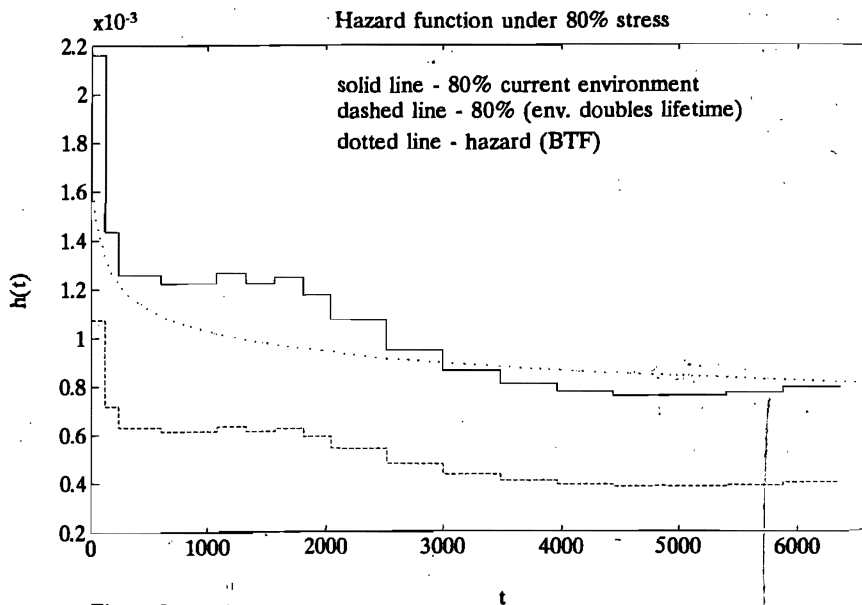


Figure 2

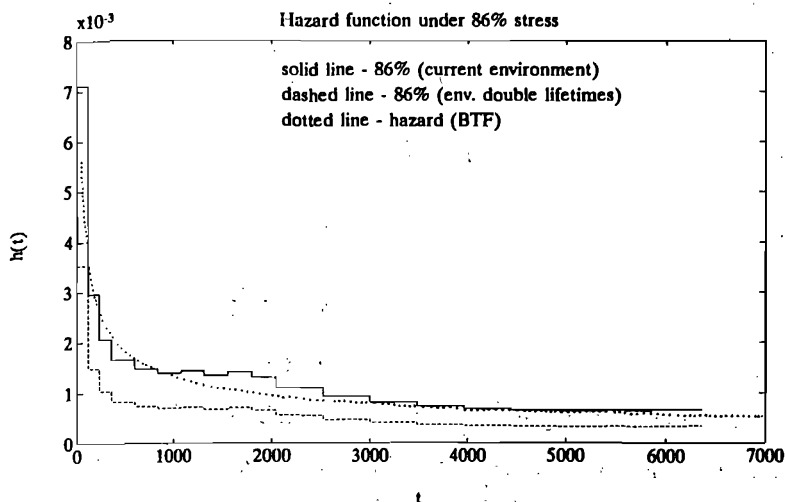


Figure 3

4. MULTI-GROUP CASE

In this section an extension of the model, discussed in section 2, to the multi-group case, is given. It is shown how the hazard functions of the lifetimes of individuals in different groups, functioning in a common environment, are estimated simultaneously and how the hazard under other environments can be obtained. The method is illustrated by application to real data.

Let $T_i = (t_{i1}, \dots, t_{in_i})$, $i = 1, 2, \dots, k$, be the survival times of individuals in k different groups in the interval $I_n = (t_{n-1}, t_n]$, $n = 1, 2, \dots, N$, where the survival time of an individual is assumed to be piecewise exponentially distributed in I_n , conditional on a common environmental effect η and failure rate λ_{ni} . The assumed relations are

$$(13) \quad \begin{aligned} \ell n \eta &= \beta_0 \\ \ell n \lambda_{ni} &= z_n^{(i)} \beta_n^{(i)'} \end{aligned}$$

where $z_n^{(i)} = (1 \quad z_n^{(i)})$ and $\beta_n^{(i)} = (\beta_0 \quad \beta_n^{(i)'})$.

The same evolution equation (3) and partially specified posterior and prior distributions, hold. The likelihood for an interval is

$$\ell n = \prod_{j=1}^{r_n} [\eta_j \lambda_{ni}^{(j)}]^{\delta_{ni}^{(j)}} \exp(-\eta_j \lambda_{ni}^{(j)} (t_{ni}^{(j)} - t_{n-1}))$$

where r_n is the total number of individuals, for the k groups, known to be alive at the beginning of I_n . $\delta_{ni}^{(j)}$ is the indicator of death for the j -th individual in group i . $t_{ni}^{(j)}$ is the survival time of the j -th individual of group i in I_n and becomes t_n if the individual survives through I_n . Sub- and superscripts are omitted for convenience.

Since updating takes place one step at a time, the observations in I_n must be ordered. If a death is observed on the j -th individual, which can be from group i , $i = 1, 2, \dots, k$, the likelihood becomes

$$(14) \quad \ell n = [\eta \lambda_i]^{\delta_i} \exp(-\eta \lambda_i (t - t_{n-1}))$$

Assume the following joint prior moments:

$$(15) \quad \begin{pmatrix} \beta \\ \ell n \eta \\ \ell n \lambda_i \end{pmatrix} \sim \left[\begin{pmatrix} a \\ f \end{pmatrix}, \begin{pmatrix} P & S' \\ S' & Q \end{pmatrix} \right]$$

where $\beta = (\beta_0 \quad \beta^{(i)'})'$, $a = (a_0 \quad a^{(i)'})'$, $P = \begin{pmatrix} p_{00} & p_{0i} \\ p_{i0} & p_{ii} \end{pmatrix}$.

It can further be seen that

$$f_0 = E(\ell n \eta) = a_0, \quad f_i = E(\ell n \lambda_i) = z^{(i)} a^{(i)},$$

$$p_{00} = \text{Var}(\beta_0), \quad p_{ii} = \text{Var}(\beta^{(i)}), \quad p_{0i} = \text{Cov}(\beta_0, \beta^{(i)}), \quad i = 1, 2, \dots, k.$$

After information D has been processed, the posterior moments of β in (15) can be obtained:

$$(16) \quad m = a + SQ^{-1}\{E^* - f\} \\ C = P - SQ^{-1}S' + SQ^{-1}V^*Q^{-1}S'$$

where
$$E^* = \begin{pmatrix} E(\ln \eta | D) \\ E(\ln \lambda_i | D) \end{pmatrix}$$

and
$$V^* = \begin{pmatrix} \text{Var}(\ln \eta | D) & \text{Cov}(\ln \eta, \ln \lambda_i | D) \\ \text{Cov}(\ln \lambda_i, \ln \eta | D) & \text{Var}(\ln \lambda_i | D) \end{pmatrix}$$

need to be determined.

Assume that the joint prior density of $\eta, \lambda_1, \dots, \lambda_k$ is a Multivariate Gamma-Beta type 2 distribution, denoted by $\text{MGB}_2(\alpha, \tau, \rho_1, \gamma_1, \dots, \rho_k, \gamma_k)$, with density function

$$(17) \quad p(\eta, \lambda_1, \dots, \lambda_k) \propto \eta^{\alpha-1} e^{-\tau\eta} \prod_{i=1}^k \lambda_i^{\rho_i-1} e^{-\gamma_i\eta\lambda_i}, \quad \eta, \lambda_i > 0.$$

It follows from (17) that $\eta, \lambda_i \sim \text{GB}_2(\alpha - \sum_{j \neq i} \rho_j, \tau, \rho_i, \gamma_i)$, with probability density function

$$(18) \quad p(\eta, \lambda_i) = \frac{\gamma_i^{\rho_i} \tau^{\alpha - \sum \rho_j}}{\Gamma(\rho_i) \Gamma(\alpha - \sum \rho_j)} \eta^{\alpha - \sum_{j \neq i} \rho_j - 1} e^{-\tau\eta} \lambda_i^{\rho_i-1} e^{-\gamma_i\eta\lambda_i}$$

By combining the likelihood (14) with the prior (18), the joint posterior distribution of $\eta, \lambda_i | D$ is found to be $\text{GB}_2(\alpha - \sum_{j \neq i} \rho_j + \delta_i, \tau, \rho_i + \delta_i, \gamma_i + \delta_i, \gamma_i + t - t_{n-1})$. The marginal posterior moments are

$$E(\eta | D) = (\alpha - \sum \rho_j) \tau^{-1}$$

$$\text{Var}(\eta | D) = (\alpha - \sum \rho_j) \tau^{-2}$$

$$E(\lambda_i | D) = \frac{\tau(\rho_i + \delta_i)}{(\gamma_i + t - t_{n-1})(\alpha - \Sigma \rho_i - 1)}$$

$$\text{Var}(\lambda_i | D) = \frac{\tau^2(\rho_i + \delta_i)(\alpha - \Sigma_{j \neq i} \rho_j + \delta_i - 1)}{(\gamma_i + t - t_{n-1})^2(\alpha - \Sigma \rho_i - 1)^2(\alpha - \Sigma \rho_i - 2)}$$

$$\text{with } \text{Cov}(\eta, \lambda_i | D) = \frac{-(\rho_i + \delta_i)}{(\gamma_i + t - t_{n-1})(\alpha - \Sigma \rho_i - 1)}, \quad \alpha - \Sigma \rho_i > 2, \tau > 0;$$

$$\gamma_i + t - t_{n-1} > 0, \quad i = 1, 2, \dots, k$$

From the prior moments, by using a Taylor series expansion, it follows that

$$(19) \quad E(\ell n \eta) = \ell n \left(\frac{\alpha - \Sigma \rho_i}{\tau} \right)$$

$$\text{Var}(\ell n \eta) = (\alpha - \Sigma \rho_i)^{-1}$$

$$E(\ell n \lambda_i) = \ell n \left(\frac{\tau \rho_i}{\gamma_i(\alpha - \Sigma \rho_i - 1)} \right)$$

$$\text{Var}(\ell n \lambda_i) = \frac{\alpha - \Sigma_{j \neq i} \rho_j - 1}{\rho_i(\alpha - \Sigma \rho_i - 2)}$$

and from the posterior moments

$$(20) \quad E(\ell n \eta | D) = \ell n \left(\frac{\alpha - \Sigma \rho_i}{\tau} \right)$$

$$\text{Var}(\ell n \eta | D) = (\alpha - \Sigma \rho_i)^{-1}$$

$$E(\ell n \lambda_i | D) = \frac{\tau(\rho_i + \delta_i)}{(\gamma_i + t - t_{n-1})(\alpha - \Sigma \rho_i - 1)}$$

$$\text{Var}(\ell n \lambda_i | D) = \frac{\alpha - \Sigma_{j \neq i} \rho_j + \delta_i - 1}{(\rho_i + \delta_i)(\alpha - \Sigma \rho_i - 2)}$$

with

$$\text{Cov}(\ell n \eta, \ell n \lambda_i | D) = -(\alpha - \Sigma \rho_i)^{-1}$$

Equations (20) can now be substituted into (16) to obtain the new prior moment of β after applying the evolution equations. By following the same procedure as in the single-group case, it is found that

$$\begin{aligned}
 (21) \quad \alpha &= p_{00}^{-1} + \Sigma \rho_i \\
 \tau &= p_{00}^{-1} \exp(-a_0) \\
 \rho_i &= \frac{1 - p_{00}}{z^{(i)} P_{ii} z^{(i)} (1 - 2p_{00}) - p_{00}} \\
 \gamma_i &= \frac{\tau \rho_i p_{00} \exp(-z^{(i)} a^{(i)})}{1 - p_{00}}
 \end{aligned}$$

Updating of the posterior moments can continue for all individuals known to be alive at the beginning of I_n , $n = 1, 2, \dots, N$ and be repeated for all intervals. The estimates obtained for β are used to estimate the hazard for the current interval before moving to the next interval.

5. ILLUSTRATION

The method of obtaining hazard functions by using the dynamic Bayes approach for multi-group data is illustrated by means of data obtained from a research report by Worku and Makatjane (1994). The study, of which the main objective was to identify factors that strongly influence child survival in Lesotho, was carried out in the rural and urban parts of the Maseru district in Lesotho during 1994. At each household, the pregnancy history of the mother over the past five years (since 1989) was recorded.

The main interest of this example is the lifetimes of children in four different groups, summarized in table 4. The numbers without brackets include all the number of deaths and survivals, while the numbers in brackets indicate the number of deaths.

From the data set (available from the authors), the age of the mother, the age of the child, whether the child was breast fed and an indication of

Table 4

Child breast fed	Age of the mother	
	Younger than 30 years	30 years or older
Yes	2144 (96)	1385 (105)
No	272 (24)	200 (56)

death are known. Table 4 shows that 281 of the 4001 live births under five years of age were infant deaths. One of the factors among the top 10 causes of child mortality in the Maseru District is the short duration of breast feeding, which indicates that the hazard rate, in the case where a child is not breast fed, is expected to be higher than in the case where the child is breast fed. A factor, listed among the top 10 causes for child morbidity in the district, is a very low or very high age of the mother.

To estimate the hazard rates of the different groups, by using the dynamic generalized Bayes method described, the following prior parameters for β were used:

$$a_{00} = (0.7 \ 0 \ 0 \ 0 \ 0)', \quad P_{00} = \begin{bmatrix} 0.001 & 0 \\ 0 & 100I_4 \end{bmatrix}$$

$$\text{From the system evolution, } W_n = \begin{bmatrix} 0.0001 & 0 \\ 0 & 0.025I_4 \end{bmatrix}$$

$$G_n = I_5, \quad w_n = (0 \ 0 \ 0 \ 0 \ 0)'$$

have been chosen and 20 intervals of length 0.25 as the time grid.

The values 0.7 and 0.001 are chosen as explained in the example of section 3 and the lack of prior information is expressed in the values of the elements of a_{00} and P_{00} .

Values of W_n , in this case 0.1 length (I_n), were set on an illustrative basis, although other choices can also be examined (Gamerman, 1987).

The estimates a_n of β_n , $n=1, 2, \dots, 20$, after smoothing had been

done, are given in table 5.

Table 5. Estimates a_n for 20 intervals

a_1	a_2	a_3	a_4	a_5
0.6953	0.6946	0.6934	0.6932	0.6919
-3.8698	-4.0223	-4.1647	-4.1399	-4.0842
-0.7178	-0.8736	-1.0194	-0.9979	-1.0446
-2.8717	-2.7872	-2.8370	-2.7803	-3.0518
-1.1760	-1.0948	-1.2638	-1.3506	-1.6213
a_6	a_7	a_8	a_9	a_{10}
0.6909	0.6908	0.6907	0.6912	0.6921
-4.1484	-4.1025	-3.9885	-3.8285	-3.6454
-1.2174	-1.2854	-1.2898	-1.2518	-1.1934
-3.1007	-3.0268	-3.0302	-2.9275	-2.7693
-1.7052	-1.6913	-1.7818	-1.7861	-1.7504
a_{11}	a_{12}	a_{13}	a_{14}	a_{15}
0.6925	0.6933	0.6945	0.6956	0.6963
-3.5657	-3.4555	-3.3217	-3.0954	-3.0996
-1.0998	-0.9828	-0.8489	-0.6227	-0.6269
-2.6591	-2.4750	-2.2378	-2.1354	-1.9131
-1.7825	-1.7560	-1.6883	-1.7760	-1.7732
a_{16}	a_{17}	a_{18}	a_{19}	a_{20}
0.6972	0.6979	0.6990	0.6998	0.7000
-3.0067	-2.9561	-2.8660	-2.7434	-2.6555
-0.5340	-0.4834	-0.3933	-0.2707	-0.1827
-1.7277	-1.5907	-1.3887	-1.2995	-1.3291
-1.8459	-2.0017	-2.1182	-2.1866	-2.2162

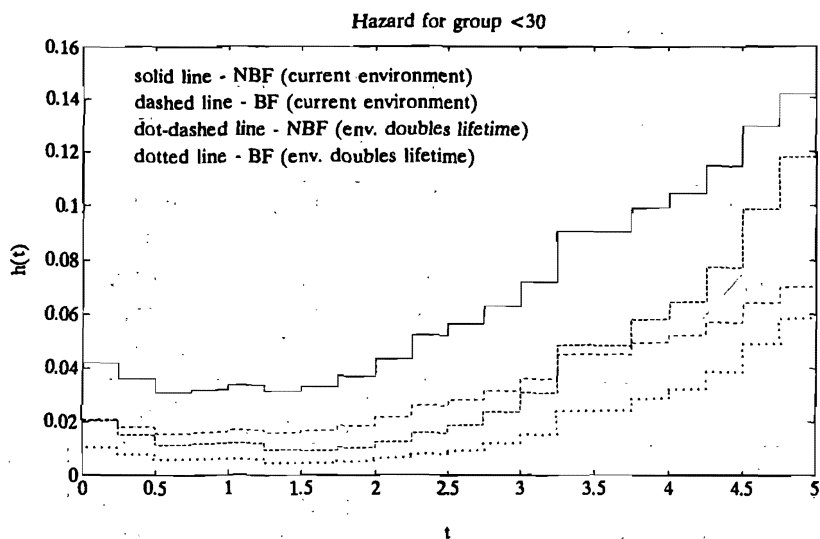


Figure 4

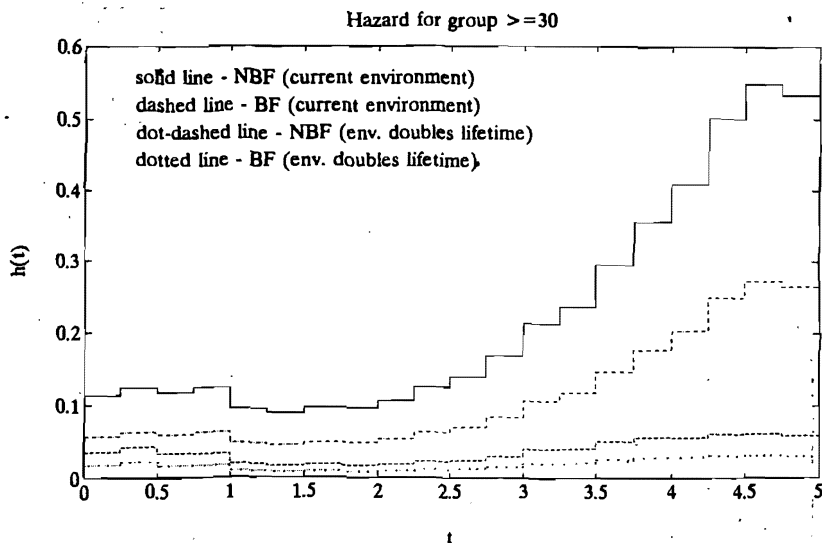


Figure 5

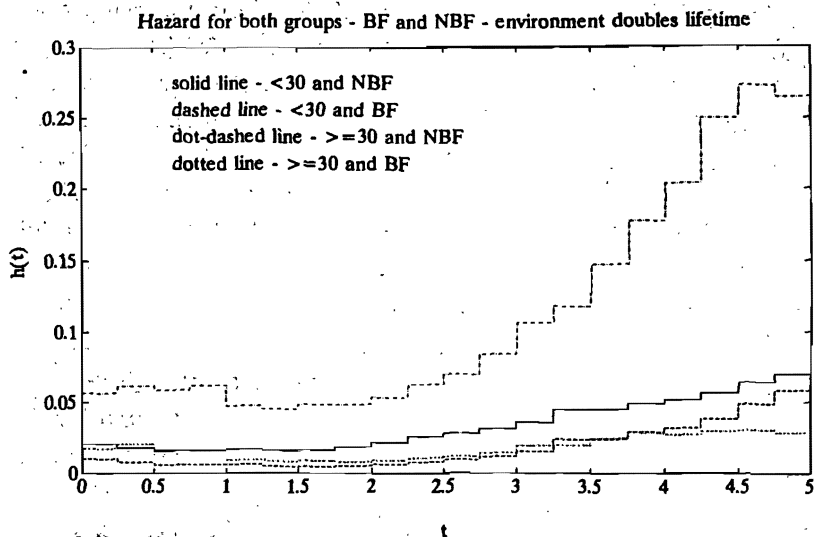


Figure 6

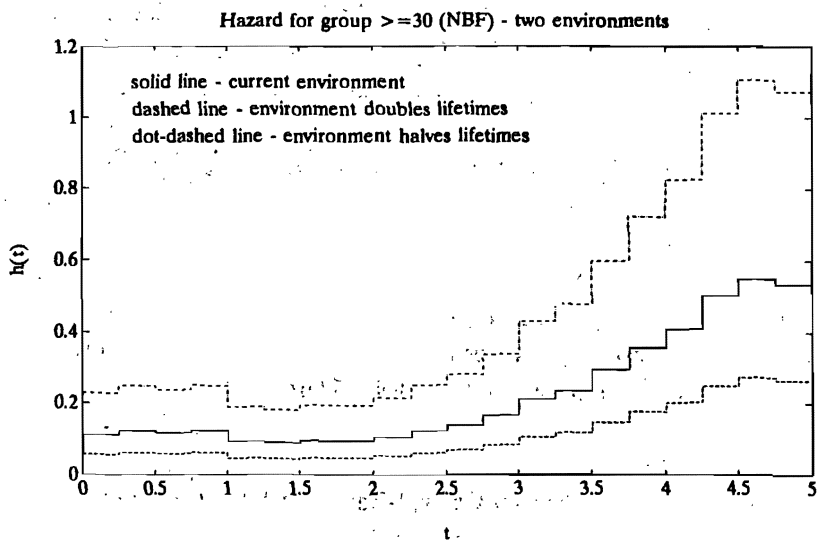


Figure 7

Figure 4 shows the hazard functions of the lifetimes of children that were not breast fed (NBF) and those who were breast fed (BF), where the mothers were of age younger than 30 years. The term "current environment" refers to the hazard function under the current conditions (as observed during the project). The graph also shows the hazard functions where the current environment is omitted, i.e. where it is assumed that other conditions can be introduced that will have a positive effect on the lifetime of the child – in this case the conditions are assumed to double the lifetime of the child. An example of such a situation is to encourage mothers to attend postnatal care health services and to make sure that infants are fully immunized.

The hazard functions, as explained for the cases shown in figure 4, can also be seen in figure 5 where the mothers were of age 30 years or older. For both mentioned groups it can be seen from the graphs that the highest hazard rate occurs where the children were not breast fed.

In figure 6 the hazard functions of the lifetimes of children for the four cases shown in table 4 can be seen, where it is assumed that certain conditions can be introduced to double the lifetimes of children. In this graph it is also noticed that the highest hazard rate is observed in cases where the mother of the child was of age 30 years or older and the child was not breast fed.

Another condition is introduced in figure 7, namely the effect on the hazard function of lifetimes is examined where it is assumed that conditions had deteriorated to such an extent, that the lifetimes of children are likely to be half that of the lifetimes observed during the research (current environment) project. An example is the presence of famine, particularly in the rural parts of the district, or an increasing number of mothers suffering from tuberculosis and passing it on to their children.

6. CONCLUSION

It was illustrated how the joint Gamma-Beta type 2 distribution can be used in a dynamic Bayes procedure to obtain the hazard rates of survival times of individuals (components) in k groups, functioning in an environment. The

model seems also useful, since it can be used to estimate the hazard function in different environments by changing the environment parameter η . Aspects that still need to be investigated more thoroughly are

- (i) the choice of the length of the interval I_n ; and
- (ii) the choice of the prior hyperparameter and matrices that appear in the evolution equations.

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